

Methods of de novo gene prediction in Prokaryotes

ORFfinder works by finding all open reading frames between start and stop codons on all frames.

Different results with different parameters: "ignore nested" reduces overlapping ORFs, allowing alternative starts increases the number of predicted ORFs.

It seems to overpredict because it treats many random ORFs as genes. ORFfinder is suitable for quick exploratory search for possible coding regions, detecting candidate ORFs around known features, but not as a final genome annotation method.

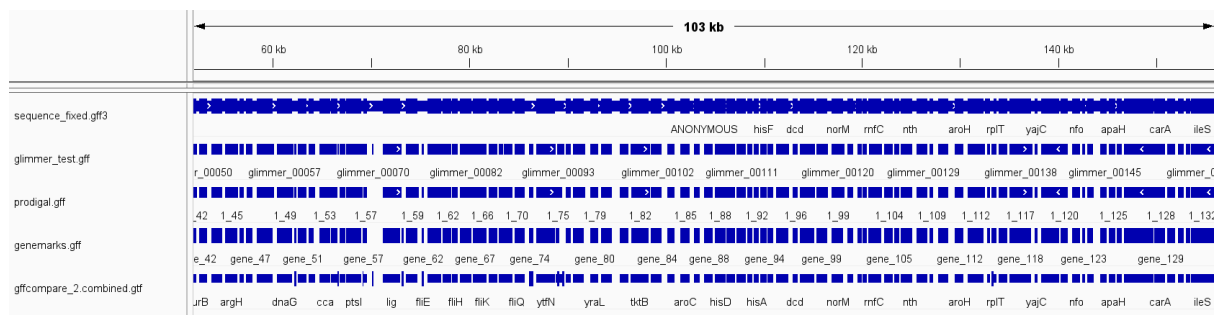
What additional information **prodigal** uses: Uses features learned from curated bacterial genomes (learned gene length distributions, coding statistics, start context and RBS signals in addition to simple ORF boundaries) and iteratively refines a model on the input genome with dynamic programming to call genes on both strands.

glimmeropts = "-l -o50 -g110 -t30"

These options control how many and what kind of ORFs Glimmer accepts as genes. (linear/circular genome, overlap of ORFs acceptable length, min ORF length, score).

The results can be improved by using a better training set, tuning overlap and length/score thresholds, incorporating biological signals or external evidence.

Tool	# predicted genes	Sensitivity vs ref	Precision vs ref	# matching transcripts
Prodigal	135	26.6%	99.4%	130
<u>Glimmer3</u>	140	26.5%	98.8%	130
GeneMarkS-2	136	26.5%	99.4%	129



Gene predictions seem to be very consistent with annotated genome.

How to improve results: Use better training sets or species-specific models to tune Markov/HMM parameters.

Incorporate additional biological signals such as RBS motifs, codon usage of known genes, or external evidence (proteins from related species, RNA-seq) when available.

In high-GC genomes, combine multiple tools and accept only genes supported by more than one, or manually inspect borderline cases.

The origin of the longest ORF is *Buchnera aphidicola* str. Bp (*Baizongia pistaciae*).

This is gene="rpoC"